

# Design and Optimization of an 11 kW Integrated OBC-LDC EV Battery Charging System

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**Abstract**—This article proposes a new integrated battery charging system (IBCS) for electric vehicles, combining a dual-active bridge (DAB) converter for the on-board charger (OBC) and a phase-shift full-bridge (PSFB) converter for the low-voltage dc–dc converter (LDC). Unlike conventional IBCSs, which employ the output inductors and transformer separately, the proposed IBCS integrates them using a two-transformer structure, thereby reducing the overall volume and the number of conducting components on the LDC secondary side. In addition, synchronous rectifiers (SRs) with an optimal driving scheme are introduced to minimize conduction losses. Furthermore, the transformers are designed to optimize the efficiency of the proposed IBCS in both grid-to-vehicle (G2V) and vehicle-to-grid (V2G) modes. These optimizations are based on a detailed analysis, including the coupling effects between the OBC and LDC and the impact of leakage inductances on the OBC primary and secondary sides, which have not been thoroughly addressed in previous studies. As a result, the proposed IBCS can achieve high power density and efficiency. Finally, to validate its performance, a prototype integrating an 11 kW OBC with a 2 kW LDC is built and tested, demonstrating a peak efficiency of 97.1% in both the G2V and V2G modes.

**Index Terms**—Electric vehicles (EVs), integration, low voltage dc–dc converter (LDC), on-board charger (OBC).

## I. INTRODUCTION

CONVENTIONAL battery charging systems (BCSs) for electric vehicles (EVs) typically consist of an on-board charger (OBC) and a low-voltage dc–dc converter (LDC) [1], [2], [3], as illustrated in Fig. 1(a). The OBC is powered by the ac grid and charges a high-voltage (HV) battery through

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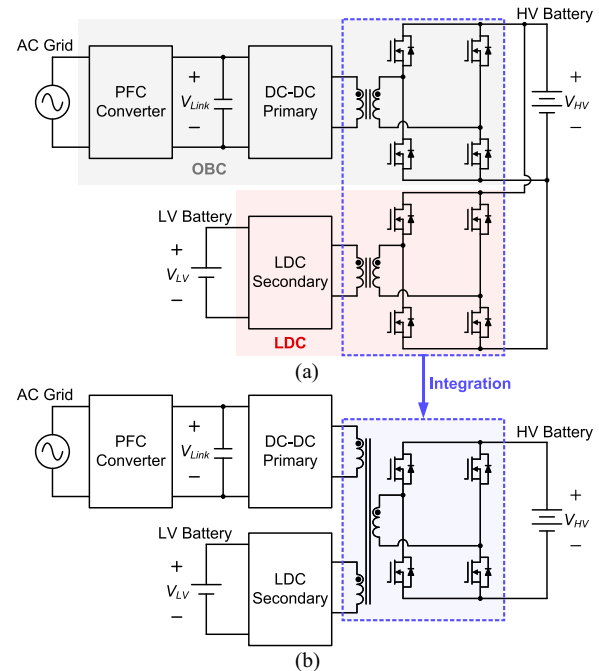


Fig. 1. Block diagram of BCS. (a) Conventional BCS. (b) Integrated BCS (IBCS).

a power factor correction (PFC) converter followed by a dc–dc converter, while the LDC is powered by the HV battery and charges a low-voltage (LV) battery [4], [5]. In each stage of dc–dc converters, galvanic isolation and efficient charging for each battery are essential. Therefore, the phase-shift full-bridge (PSFB) converter, LLC converter, and dual-active bridge (DAB) converter have been preferred for the OBC [6], [7], [8]. For the LDC, to achieve a large step-down ratio and low rms current, the PSFB converter has been widely utilized [9], [10].

Recently, BCSs are required to achieve high power density, low cost, and high efficiency because they are installed in EVs with limited space and significantly affect the battery charging time. In particular, this trend has spurred extensive research on integrated OBC-LDC BCSs (IBCSs) [11], [12], [13], [14], [15], [16]. The IBCSs are implemented by merging the switches on the OBC secondary and LDC primary sides, which share a common ground and the same structure. Additionally, the transformers of the OBC and LDC can be combined using a multiwinding transformer, as illustrated in Fig. 1(b). This

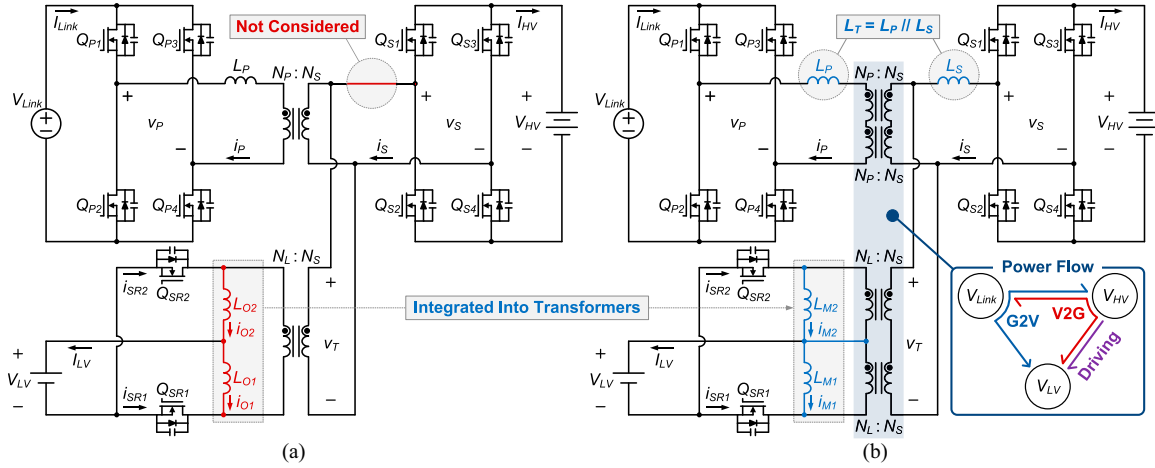


Fig. 2. Circuit diagram. (a) Conventional IBCS [16]. (b) Proposed IBCS.

integration effectively improves the power density and lowers the overall cost by reducing the total number of components.

Meanwhile, IBCSs should be able to perform the following three operational modes: grid-to-vehicle (G2V) mode, vehicle-to-grid (V2G) mode, and driving mode. To satisfy this requirement, bidirectional power transfer capability and at least two control variables for two independently regulated output ports, e.g., HV and LV batteries in the G2V mode, are essential.

Therefore, various types of IBCSs equipped with two control variables have been proposed [11], [12], [13], [14], [15], [16]. In [11], the IBCS was implemented by integrating two PSFB converters for the OBC and LDC, while configuring a built-in buck converter on the LDC secondary side with an additional switch and diode. By regulating the H-bridges and buck switch to control the OBC and LDC, respectively, this approach achieved independent battery charging. However, the additional components can cause high conduction and switching losses. In [12] and [13], the LLC converter for the OBC was combined with the PSFB converter. Since they applied pulse-frequency modulation in the G2V and V2G modes, decoupling switching cells were employed on the LDC secondary side to achieve two control variables. Therefore, similar to [11], the IBCSs in [12] and [13] met all operating requirements at the expense of additional power losses. In [14], only two switches for the LLC and PSFB converters were partially merged to configure two H-bridges on the OBC secondary side, sharing a common leg. By utilizing these H-bridges for regulating each port, the batteries were charged independently. However, partial integration of the switches and the use of individual transformers could limit improvements in power density and cost. Finally, in [15] and [16], the DAB converter for the OBC was integrated with the PSFB converter, as illustrated in Fig. 2(a). By utilizing two control variables in H-bridges to independently regulate the outputs for the OBC and LDC, [15] and [16] implemented IBCSs without requiring additional switches or diodes, thereby significantly improving power density and efficiency.

Meanwhile, conventional IBCSs exhibit several limitations in performance optimization and have not yet been validated

under high-power conditions. First, while previous studies employed a center-tapped rectifier [11], [12], [13], a current-doubler rectifier [14], [16], and a full-bridge rectifier [15], the overall volume and the number of conducting components on the LDC secondary side can be further reduced by adopting a two-transformer structure [10]. Second, conventional IBCSs were not optimized for both the G2V and V2G modes. In [11], [12], [13], [14], this limitation is due to the inherent asymmetric characteristics of PSFB and LLC converters. In [15] and [16], it is attributed to the coupling effects induced by the transformer integration, which strongly influence bidirectional performance and are affected by the transformer leakage inductances. However, as shown in Fig. 2(a), the inductance on the secondary side was not considered, despite its significant impact on bidirectional performance. Third, to reduce conduction losses in the LDC, synchronous rectifiers (SRs) were employed in [16], but they were synchronized with the lagging-leg switches of either the OBC primary or secondary side. This can cause body diode conduction in the SRs during zero-voltage intervals of both H-bridges, thereby increasing conduction losses. Finally, although the global trend is moving toward high-power battery chargers to shorten charging times [4], previous studies demonstrated the effectiveness of IBCSs only below 3.7 kW. Since the influence of the coupling effects on bidirectional performance increases with the power level, further research is essential for the optimal design of high-power bidirectional chargers.

To address these issues, this article proposes a new IBCS that integrates the DAB converter for the OBC and a two-transformer PSFB converter with the SRs for the LDC. As shown in Fig. 2(b), the proposed IBCS incorporates the output inductors into the transformers, thereby improving power density and reducing the number of conducting components on the LDC secondary side compared to conventional IBCSs [10].

Table I presents a comparison with previous studies, including topological advantages of the proposed IBCS and key contributions of this article. The contributions are as follows.

- 1) A detailed analysis of the proposed IBCS is presented. This includes the coupling effects between the OBC and

TABLE I  
COMPARISON WITH PREVIOUS STUDIES

|                                         | [11]          | [12]            | [13]            | [14]          | [15]            | [16]                         | Prop.            |
|-----------------------------------------|---------------|-----------------|-----------------|---------------|-----------------|------------------------------|------------------|
| Topology (OBC / LDC)                    | PSFB / PSFB   |                 | LLC / PSFB      |               | DAB / PSFB      |                              | DAB / PSFB       |
| LDC Rectifier Type                      | Center-Tapped |                 | Current-Doubler |               | Full-Bridge     | Current-Doubler              | Two-Transformer  |
| Conducting Components*                  | 4             |                 | 3               |               | 4               | 3                            | 2                |
| Coupling Effects Between OBC and LDC    | Considered    |                 | Not Considered  |               |                 |                              | Considered       |
| Design Optimization in Both G2V and V2G |               |                 |                 | X             |                 |                              | O                |
| SR Driving Scheme                       | -             |                 | Not Specified   |               | -               | Sync w/ Lagging-Leg Switches | Optimized Method |
| Output Power (Total / LDC)              | 3.3 kW / 1 kW | 3.3 kW / 1.8 kW | - / 1 kW        | 3.7 kW / 1 kW | 3.5 kW / 0.5 kW | 1.5 kW / 0.3 kW              | 11 kW / 2 kW     |

Note: \*Conducting Components: Total Number of Conducting Semiconductor Devices and Windings in Magnetic Components on the LDC Secondary Side.

LDC, as well as the influence of the leakage inductances on the OBC primary and secondary sides in the G2V and V2G modes, which have not been thoroughly explored in previous studies.

- 2) Design guidance for the transformer parameters is provided to simultaneously optimize performance in both the G2V and V2G modes, while satisfying all operational requirements.
- 3) An optimal SR driving scheme based on an OR-gated method is proposed to minimize conduction losses in the LDC.
- 4) The effectiveness and feasibility of the proposed IBCS are demonstrated using an 11 kW prototype, offering insights into its application to high-power battery chargers.

## II. CONTROL SCHEME AND OPERATIONAL PRINCIPLES

As illustrated in Fig. 2(b), the proposed IBCS has three operational modes: G2V mode, V2G mode, and driving mode. In this section, the control schemes and operational principles for each mode are presented.

### A. Control Scheme

The IBCS requires two control variables to independently regulate the outputs of the OBC and LDC. Therefore, dual phase-shift (DPS) control is adopted in this article [17]. In DPS control, the phase-shift ratio between the H-bridges,  $D_2$ , is adjusted to regulate power transfer, while the duty ratio of the zero-voltage,  $D_1$ , is used to control the average voltage of the transformer. Accordingly,  $D_2$  and  $D_1$  are utilized to control the OBC and LDC, respectively.

Considering the power flow, shown in Fig. 2(b), each control variable is adjusted as follows. 1)  $D_1 \geq 0$  and  $D_2 \geq 0$  in the G2V mode, 2)  $D_1 \geq 0$  and  $D_2 \leq 0$  in the V2G mode, and 3)  $D_1 \geq 0$  and  $D_2 = 0$  in the driving mode. Due to the symmetry in bidirectional operation, the V2G mode is analyzed based on the G2V mode. Figs. 3(a) and 4 illustrate the key waveforms and equivalent circuits in the G2V mode, while Figs. 3(b) and 5 correspond to the driving mode. To simplify the analysis, several assumptions are made as follows.

- 1) The components that are not marked in Fig. 2(b) are ignored.
- 2) The dead-time between the gate signals of the switches  $Q_{P1}$ – $Q_{P4}$  and  $Q_{S1}$ – $Q_{S4}$  is small enough to be negligible.
- 3) Transformers have a turns ratio  $n$ , defined as  $n = N_p / N_L = N_s / N_L$ , where  $N_p$ ,  $N_s$ , and  $N_L$  are the turn numbers in each port.
- 4) The magnetizing inductances of transformers are identical, i.e.,  $L_{M1} = L_{M2} = L_M$ .
- 5) The dc-link voltage  $V_{Link}$  and the battery voltages  $V_{HV}$  and  $V_{LV}$  are constant during a switching period  $T_s$ .
- 6) In the driving mode, although the OBC primary side is under no-load conditions,  $Q_{P1}$ – $Q_{P4}$  are turned ON to prevent resonance between their output capacitances and the transformer leakage inductances, which can cause an excessive rise in  $V_{Link}$  [12]. Thus, only a small circulating current flows through this port, which is assumed to be negligible.

### B. Operational Principles in G2V Mode

As shown in Fig. 3(a), the switching cycle in the G2V mode is divided into two half-cycle intervals,  $t_0$ – $t_5$  and  $t_5$ – $t_{10}$ . Due to the symmetry, only the first half-cycle is explained.

**State 1 [ $t_0$ – $t_1$ ]:** State 1 begins as  $V_{Link}$  is applied to the primary side. The OBC primary current  $i_P$  increases, and the secondary side of the LDC starts commutation. In this state, since short circuits are formed on both secondary sides of the OBC and LDC, the OBC secondary current  $i_S$  remains constant, as shown in Figs. 3(a) and 4(a). In the two-transformer PSFB converter,  $L_{M1}$  and  $L_{M2}$  have an offset current of  $I_{LV}/2$  [10], where  $I_{LV}$  is the LDC output current. Thus, when ignoring the current ripple on  $L_{M1}$  and  $L_{M2}$ , the duration of this state is obtained as follows:

$$t_{01} = D_{C2} T_{hs} = \frac{L_P I_{LV}}{n V_{Link}} \quad (1)$$

where  $D_{C2}$  denotes the duty ratio for commutation in the G2V mode,  $T_{hs}$  is half of the switching period, and  $L_P$  is the leakage inductance on the OBC primary side.

**State 2 [ $t_1$ – $t_4$ ]:** After commutation, the difference between  $i_P$  and  $i_S$  is reflected on the LDC secondary side, which is the same

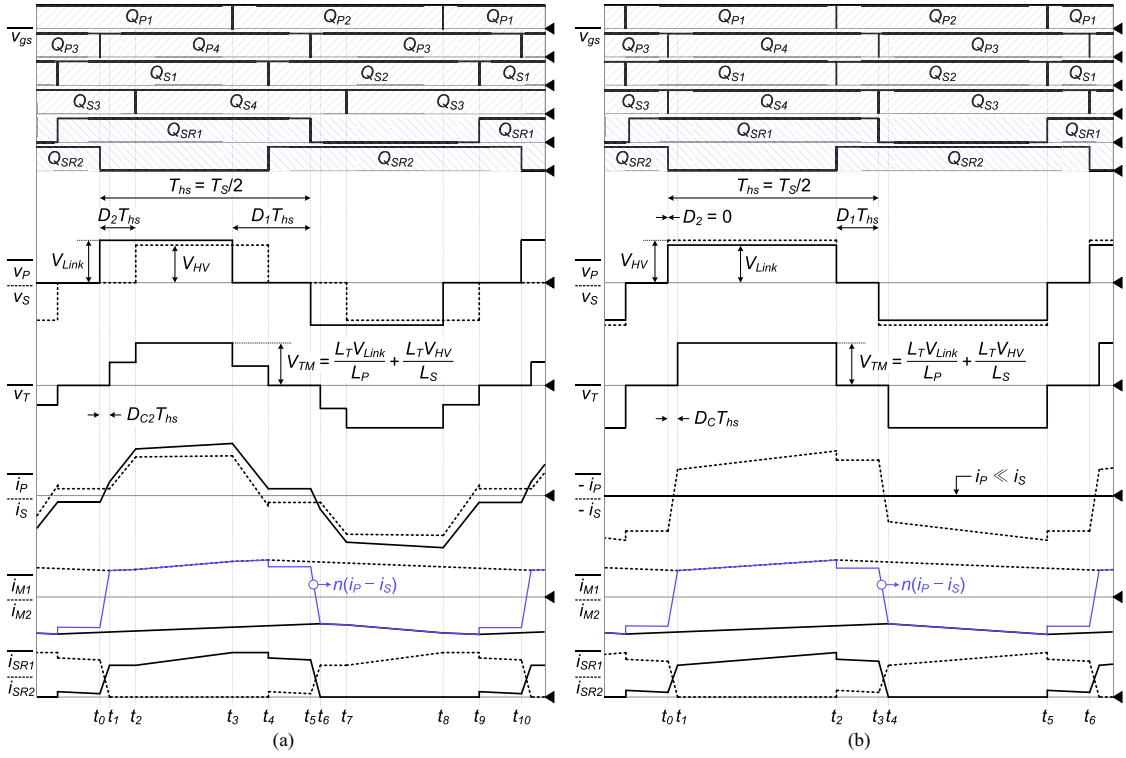


Fig. 3. Key waveforms of proposed IBCS. (a) G2V mode. (b) Driving mode.

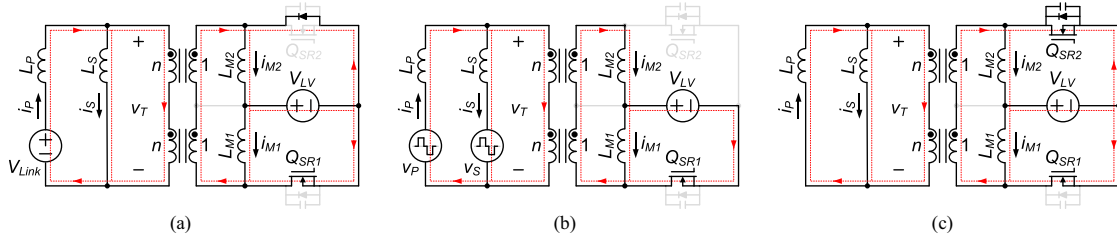


Fig. 4. Equivalent circuits in G2V mode. (a) State 1  $[t_0-t_1]$ . (b) State 2  $[t_1-t_4]$ . (c) State 3  $[t_4-t_5]$ .

with the magnetizing current  $i_{M2}$ , as illustrated in Figs. 3(a) and 4(b). Therefore, the following equation can be obtained based on the Kirchhoff's current law:

$$\frac{v_P(t) - v_T(t)}{L_P} - \frac{v_T(t) - v_S(t)}{L_S} = \frac{v_T(t) - nV_{LV}}{n^2L_M} \quad (2)$$

where  $v_P$  and  $v_S$  are the terminal voltages of the H-bridges,  $L_S$  is the leakage inductance on the OBC secondary side, and  $v_T$  is the voltage across the transformer. Assuming  $n^2L_M$  is sufficiently large compared to  $L_P$  and  $L_S$ , (2) can be rewritten as follows:

$$v_T(t) = \frac{L_T v_P(t)}{L_P} + \frac{L_T v_S(t)}{L_S} \quad (3)$$

where  $L_T$  is the parallel inductance of  $L_P$  and  $L_S$ , as presented in Fig. 2(b). By substituting the corresponding  $v_P$  and  $v_S$  into (3),  $v_T$  during  $t_1-t_4$  can be defined. As illustrated in Fig. 4(b), the differences between the terminal voltages and  $v_T$  are applied to  $L_P$  and  $L_S$ . Thus,  $i_P$  and  $i_S$  can be expressed as follows:

$$\frac{di_P(t)}{dt} = \frac{v_P(t) - v_T(t)}{L_P} \quad (4)$$

$$\frac{di_S(t)}{dt} = \frac{v_T(t) - v_S(t)}{L_S}. \quad (5)$$

**State 3  $[t_4-t_5]$ :** State 3 begins when zero voltage is applied to both H-bridges. As illustrated in Figs. 3(a) and 4(c),  $v_T$  becomes zero, and the output capacitance of  $Q_{SR2}$  is discharged as  $n(i_P - i_S)$  decreases. Since the difference between  $n(i_P - i_S)$  and  $i_{M2}$  flows through  $Q_{SR2}$ , both SRs are turned ON to minimize conduction losses by preventing body diode conduction. The detailed control schemes for the SRs are discussed in Section III. During this state,  $i_P$  and  $i_S$  remain constant, and  $i_{M2}$  decreases with a slope of  $-V_{LV}/L_{M2}$ . This state ends at  $t_5$ , and the subsequent operations are identical to those in the previous states, except for the direction of the current.

### C. Operational Principles in Driving Mode

As shown in Fig. 3(b), the switching cycle in the driving mode is divided into two half-cycle intervals,  $t_0-t_3$  and  $t_3-t_6$ . Due to the symmetry, only the first half-cycle is explained. In

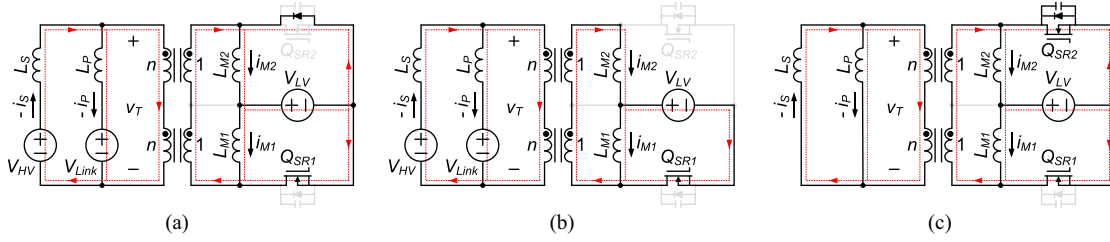


Fig. 5. Equivalent circuits in driving mode. (a) State 1  $[t_0-t_1]$ . (b) State 2  $[t_1-t_2]$ . (c) State 3  $[t_2-t_3]$ .

addition, considering the direction of power flow, as depicted in Fig. 2(b), the polarities of  $i_P$  and  $i_S$  are plotted in reverse.

**State 1  $[t_0-t_1]$ :** At time  $t_0$ , both  $V_{Link}$  and  $V_{HV}$  are applied to the H-bridges. As a result, both voltage sources participate in the commutation, and the duration of this state can be derived based on the equivalent circuit shown in Fig. 5(a) as follows:

$$t_{01} = D_C T_{hs} = \frac{I_{LV}}{n} \left( \frac{L_P L_S}{L_S V_{Link} + L_P V_{HV}} \right) \quad (6)$$

where  $D_C$  is the duty ratio for commutation in the driving mode.

**State 2  $[t_1-t_2]$ :** After commutation, the equivalent circuit becomes identical to State 2 in the G2V mode, as shown in Figs. 4(b) and 5(b). Hence, this state can be analyzed using (3)–(5).

In the conventional DAB converter,  $V_{Link}$  becomes  $V_{HV}$  when  $D_2 = 0$  [18]. However, in the proposed IBCS,  $V_{Link}$  participates in commutation, which transfers energy to  $L_P$ . Thus, during this state, the transferred energy must be compensated to satisfy the charge balance. To achieve this,  $V_{Link}$  becomes slightly lower than  $V_{HV}$  and varies with  $I_{LV}$ , as illustrated in Fig. 3(b) [12].

**State 3  $[t_2-t_3]$ :** At  $t_2$ ,  $v_P$  and  $v_S$  become zero, and both SRs are turned ON, as illustrated in Figs. 3(b) and 5(c). Therefore,  $i_P$  and  $i_S$  remain constant. This state ends at  $t_3$ , and the subsequent operations are identical to those of the previous states.

### III. ANALYSIS

This section presents the optimal SR driving scheme, the voltage gain of the LDC, and the power transfer of the OBC. Based on this analysis, operational constraints, power losses, coupling effects, and the influence of the transformer leakage inductances in bidirectional operation are discussed. The specifications used in the analysis are summarized in Table II. Fig. 6(a) and (b) show the  $V_{Link}$  profile and power distribution in the G2V and V2G modes, respectively. As presented in Fig. 6(a), when  $V_{HV} \geq 650$  V,  $V_{Link}$  is adjusted to achieve unity gain in the OBC, thereby optimizing efficiency [14]. Across the entire  $V_{HV}$  range, the total output power,  $P_{Total}$ , is regulated to its maximum value, i.e., 11 kW. To satisfy this condition, the output power of the OBC,  $P_{OBC}$ , varies with the output power of the LDC,  $P_{LDC}$ , as shown in Fig. 6(b).

#### A. Optimal SR Driving Scheme

As illustrated in Figs. 3–5, when  $v_T$  becomes zero in State 3, the output capacitances of SRs are discharged, allowing their body diodes to conduct. Therefore, to minimize the conduction

TABLE II  
SYSTEM SPECIFICATIONS

| Operational Mode | Parameters                   | Value      |
|------------------|------------------------------|------------|
| G2V and V2G mode | Output power (total/LDC)     | 11 kW/2 kW |
|                  | DC-link voltage $V_{Link}$   | 650–800 V  |
|                  | HV battery voltage $V_{HV}$  | 500–800 V  |
|                  | LV battery voltage $V_{LV}$  | 14 V       |
|                  | Switching frequency $F_{sw}$ | 100 kHz    |
| Driving mode     | Output power (LDC)           | 2 kW       |
|                  | HV battery voltage $V_{HV}$  | 500–800 V  |
|                  | LV battery voltage $V_{LV}$  | 14 V       |
|                  | Switching frequency $F_{sw}$ | 100 kHz    |

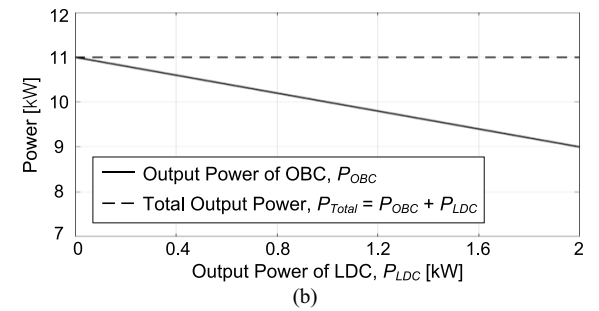
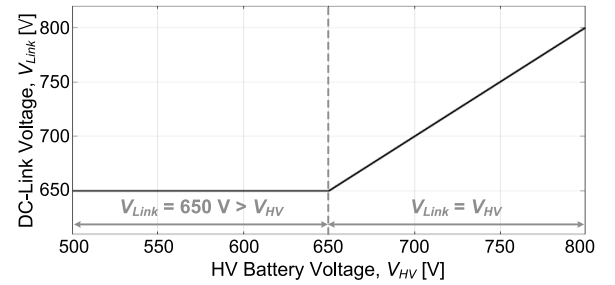


Fig. 6.  $V_{Link}$  profile and power distribution in G2V and V2G modes. (a)  $V_{Link}$  profile according to  $V_{HV}$ . (b) Power distribution.

losses, both SRs should be turned ON during all zero-voltage intervals of  $v_T$ . To address this feature, this article proposes an optimal SR driving scheme. Fig. 7 illustrates the gate signals for the conventional and proposed methods in the G2V mode under the condition  $D_1 \geq D_2$ . In [16], the SRs are synchronized with the lagging-leg switches of the leading port. As a result, the body diode of either  $Q_{SR1}$  or  $Q_{SR2}$  conducts during the zero-voltage intervals, increasing conduction losses. Conversely, the proposed method generates the gate signals for the SRs by

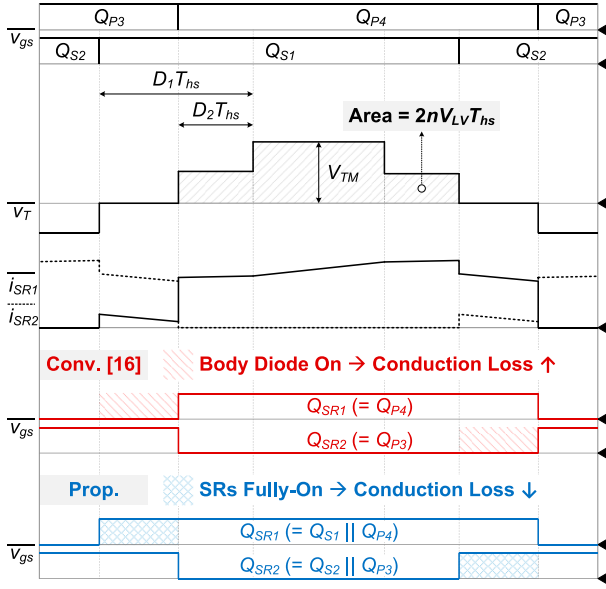


Fig. 7. Gate signals for SRs of conventional and proposed methods in G2V mode under the condition  $D_1 \geq D_2$ .

logically OR-ing the leading-leg switches of the lagging port and the lagging-leg switches of the leading port, i.e.,  $Q_{SR1} = Q_{P4} \parallel Q_{S1}$  and  $Q_{SR2} = Q_{P3} \parallel Q_{S2}$ . Consequently, both SRs are turned ON during all zero-voltage intervals, thereby preventing body diode conduction and minimizing conduction losses in the LDC.

Meanwhile, as indicated in Fig. 7, the optimal gate signals for the SRs can be defined based on  $v_T$  as follows.  $Q_{SR1}$  turns ON when  $v_T \geq 0$ , and  $Q_{SR2}$  turns ON when  $v_T \leq 0$ . This criterion remains valid regardless of the relationship between  $D_1$  and  $D_2$ . As shown in Fig. 8, when  $D_2 > D_1$ , there are no zero-voltage intervals, and the envelope of  $v_T$  varies depending on the relationship between  $V_{Link}$  and  $V_{HV}$ . Therefore, unlike the case of  $D_1 \geq D_2$ , where a unified control scheme can be applied, this condition requires distinct SR control schemes, which vary with the relationship between  $V_{Link}$  and  $V_{HV}$ . As a result, if the proposed IBCS operates under both  $D_1 \geq D_2$  and  $D_2 > D_1$  conditions, control complexity increases significantly. To address this issue, the operational constraint in the G2V mode is defined as  $D_1 \geq D_2$ , and the transformer parameters are designed accordingly in Section IV.

Table III summarizes the operational constraints and the proposed SR driving schemes for each mode. In the V2G and driving modes, the constraints and SR driving schemes are defined in the same manner as in the G2V mode, considering the polarity of  $D_2$  and the power flow. Based on these results, the proposed IBCS can minimize conduction losses in the LDC without increasing control complexity.

### B. Voltage Gain of LDC

In the proposed IBCS, the voltage gain of the LDC can be derived by applying the volt-second balance to  $L_{M1}$  or  $L_{M2}$ . As illustrated in Figs. 3(a) and 4,  $v_T/n - V_O$  is applied to  $L_{M2}$  during State 2, while  $-V_O$  is applied during the remaining intervals.

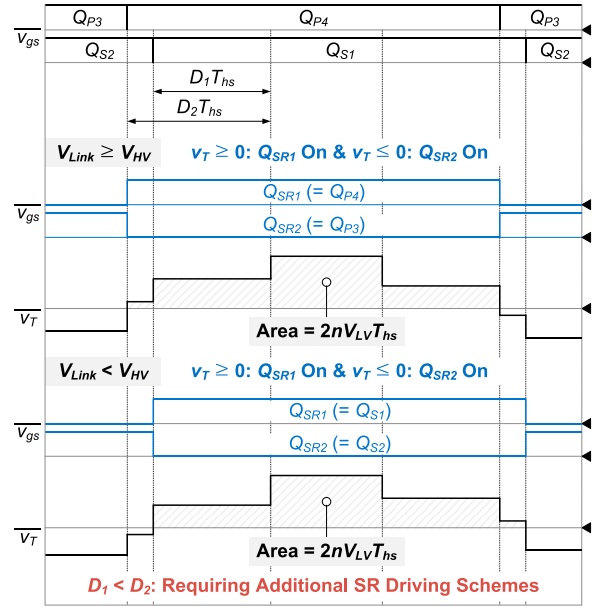


Fig. 8. Optimal gate signals for SRs based on relationship between  $V_{Link}$  and  $V_{HV}$  in G2V mode under condition  $D_2 > D_1$ .

TABLE III  
OPERATIONAL CONSTRAINTS AND PROPOSED SR DRIVING SCHEMES

| Operational Mode  | G2V                    | V2G                                                    | Driving                                                |
|-------------------|------------------------|--------------------------------------------------------|--------------------------------------------------------|
| Constraints       | $D_1 \geq D_2$         | $D_1 \geq  D_2 $                                       | $D_1 \geq 0, D_2 = 0$                                  |
| SR Driving Scheme | $Q_{SR1}$<br>$Q_{SR2}$ | $Q_{P4} \parallel Q_{S1}$<br>$Q_{P3} \parallel Q_{S2}$ | $Q_{S4} \parallel Q_{P1}$<br>$Q_{S3} \parallel Q_{P2}$ |

Therefore, by integrating  $v_T$  over  $T_{hs}$ , which corresponds to the gray-shaded area in Fig. 7, the voltage gain can be obtained as follows:

$$V_{LV} = \frac{(1 - D_1)(L_S V_{Link} + L_P V_{HV})L_T}{2nL_P L_S} = \frac{(1 - D_1)V_{TM}}{2n} \quad (7)$$

where  $V_{TM}$  represents  $v_T$  when  $V_{Link}$  and  $V_{HV}$  are applied to the H-bridges, as illustrated in Fig. 3. A similar procedure can also be applied to the case when  $D_1 < D_2$ , yielding the following equations:

$$V_{LV} = \frac{(1 - D_1)V_{TM}}{2n} - \frac{(D_2 - D_1)L_T V_{HV}}{nL_S}, \text{ for } V_{Link} \geq V_{HV} \quad (8)$$

$$V_{LV} = \frac{(1 - D_1)V_{TM}}{2n} - \frac{(D_2 - D_1)L_T V_{Link}}{nL_P}, \text{ for } V_{Link} < V_{HV}. \quad (9)$$

As indicated in (8) and (9), when  $D_1 < D_2$ ,  $V_{LV}$  is affected by the operation of the OBC, i.e.,  $D_2$ , which can degrade the control dynamics and stability of the LDC. This characteristic supports the validity of the defined operational constraints shown in Table III.

Meanwhile, the commutation period,  $D_{C2}T_{hs}$ , is not reflected in Fig. 7. To account for this, the practical voltage gain of the

TABLE IV  
CURRENT AND POWER MODELS IN G2V MODE

| Variable                          | Model                                                                                                                             | Variable              | Model                                                                                                |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------|
| $i_p(t_0)$                        | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{HV}-V_{Link})+D_{C2}V_{Link}}{L_P+L_S} - \frac{D_{C2}V_{Link}}{L_P} \right]$            | $i_s(t_0)$            | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{HV}-V_{Link})+D_{C2}V_{Link}}{L_P+L_S} \right]$            |
| $i_p(t_1)$                        | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{HV}-V_{Link})+D_{C2}V_{Link}}{L_P+L_S} + \frac{D_{C2}V_{Link}}{L_P} \right]$            | $i_s(t_1)$            | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{HV}-V_{Link})+D_{C2}V_{Link}}{L_P+L_S} \right]$            |
| $i_p(t_2)$                        | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{HV}-V_{Link})+(2D_2-D_{C2})V_{Link}}{L_P+L_S} + \frac{D_{C2}V_{Link}}{L_P} \right]$     | $i_s(t_2)$            | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{HV}-V_{Link})+(2D_2-D_{C2})V_{Link}}{L_P+L_S} \right]$     |
| $i_p(t_3)$                        | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{Link}-V_{HV})+2D_2V_{HV}-D_{C2}V_{Link}}{L_P+L_S} + \frac{D_{C2}V_{Link}}{L_P} \right]$ | $i_s(t_3)$            | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{Link}-V_{HV})-D_{C2}V_{Link}+2D_2V_{HV}}{L_P+L_S} \right]$ |
| $i_p(t_4) = i_p(t_5)$             | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{Link}-V_{HV})-D_{C2}V_{Link}}{L_P+L_S} + \frac{D_{C2}V_{Link}}{L_P} \right]$            | $i_s(t_4) = i_s(t_5)$ | $\frac{T_{hs}}{2} \left[ \frac{(1-D_1)(V_{Link}-V_{HV})-D_{C2}V_{Link}}{L_P+L_S} \right]$            |
| $P_{Total} (= P_{OBC} + P_{LDC})$ | $\frac{V_{Link}T_{hs}}{2} \left[ \frac{V_{HV}D_2(2-2D_1-D_2)}{L_P+L_S} + \frac{V_{Link}L_T D_{C2}(1-D_1-D_{C2})}{L_P^2} \right]$  | $P_{OBC}$             | $\frac{V_{Link}V_{HV}T_{hs}}{2} \left[ \frac{(1-D_1)(2D_2-D_{C2})-D_2^2}{L_P+L_S} \right]$           |

LDC is derived by incorporating the commutation period into (7) as follows:

$$V_{LV} = \frac{(1-D_1)V_{TM}}{2n} - \frac{D_{C2}L_TV_{Link}}{2nL_P} = \frac{(1-D_1)V_{TM}}{2n} - \frac{L_TI_{LV}}{2n^2T_{hs}}. \quad (10)$$

Since the obtained voltage gain is independent of  $D_2$ , both the V2G and driving modes can also be modeled using (10).

### C. Power Transfer of OBC

In the proposed IBCS, the power models,  $P_{Total}$  and  $P_{OBC}$ , in the G2V mode can be derived as follows:

$$P_{Total} = V_{Link}I_{Link} = \frac{V_{Link} \int_{t_0}^{t_3} i_p(t)dt}{T_{hs}} \quad (11)$$

$$P_{OBC} = V_{HV}I_{HV} = \frac{V_{HV} \int_{t_2}^{t_4} i_s(t)dt}{T_{hs}} \quad (12)$$

where  $i_p$  and  $i_s$  from  $t_0$  to  $t_5$ , as illustrated in Fig. 3(a), can be obtained by integrating (4) and (5) over  $T_{hs}$  [17], respectively.

The derived current and power models in the G2V mode are summarized in Table IV, where the corresponding models for the V2G mode, including  $D_{C2}$ , can be obtained by interchanging the input and output ports of the OBC.

### D. Power Loss Analysis

To quantitatively evaluate the performance of the proposed IBCS, a detailed power loss analysis was conducted. The loss models used in the analysis are presented in Table V, covering losses in the OBC switches, transformers, and SRs.

The switching losses in the OBC switches are calculated based on the overlap between voltage and current during the turn-ON and turn-OFF intervals. These losses are expressed as a function of the switching frequency  $F_{SW}$ ; turn-ON and turn-OFF times,  $t_{ON}$  and  $t_{OFF}$ ; the corresponding switching currents,  $I_{ON}$  and  $I_{OFF}$ ; and the applied voltages,  $V_{Link}$  and  $V_{HV}$ . The conduction losses are derived based on the rms currents and the MOSFET on-resistance  $R_{ds(ON)}$ . The transformer losses include core and copper losses. The core losses are modeled as a function of the switching frequency, magnetic flux swing  $\Delta B$ , effective core volume  $V_e$ , and core loss coefficients  $m$ ,  $\alpha$ , and

TABLE V  
POWER LOSS MODELS

| Component                                    | Model                                                                                                                |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| OBC primary switches ( $Q_{P1} - Q_{P4}$ )   | $\frac{V_{Link}F_{SW}}{2} (I_{on}t_{on} + I_{off}t_{off}) + R_{ds(on)} \left( \frac{i_{P(rms)}}{\sqrt{2}} \right)^2$ |
| OBC secondary switches ( $Q_{S1} - Q_{S4}$ ) | $\frac{V_{HV}F_{SW}}{2} (I_{on}t_{on} + I_{off}t_{off}) + R_{ds(on)} \left( \frac{i_{S(rms)}}{\sqrt{2}} \right)^2$   |
| Transformers                                 | $mF_{SW}^\alpha (\Delta B)^\beta V_e + R_P i_{P(rms)}^2 + R_S i_{S(rms)}^2 + R_L i_{SR(rms)}^2$                      |
| SRs ( $Q_{SR1}, Q_{SR2}$ )                   | $R_{ds(on)} i_{SR(rms)}^2$                                                                                           |

$\beta$ . The copper losses are obtained using the rms currents and winding resistances of  $N_P$ ,  $N_S$ , and  $N_L$ , i.e.,  $R_P$ ,  $R_S$ , and  $R_L$ . Finally, the conduction losses of the SRs are calculated based on the rms current and their corresponding on-resistance.

### E. Coupling Effects and Optimization of Inductance Ratio Between $L_P$ and $L_S$

In the proposed IBCS, the performance in both the G2V and V2G modes is significantly influenced by the coupling effects and the inductance ratio of  $L_P$  and  $L_S$ . The coupling effects, induced by the integration of the transformers for the OBC and LDC, refer to the interaction between their control variables, i.e.,  $D_2$  and  $D_1$ . As indicated in (1) and (10),  $D_{C2}$  and  $D_1$  vary with the LDC output power,  $P_{LDC}$ , and this variation alters the envelopes of  $i_p$  and  $i_s$  by changing  $D_2$ , affecting both switching and conduction losses in the OBC. Meanwhile, the inductance distribution, quantified by the ratio  $k = L_T/L_P = 1 - L_T/L_S$ , directly affects the envelopes of  $i_p$  and  $i_s$  during bidirectional operation. This is because  $D_{C2}$  is inversely proportional to  $L_P$  and  $L_S$  in the G2V and V2G modes, respectively. Consequently, when  $L_P \neq L_S$ , the proposed IBCS exhibits asymmetric loss distribution in bidirectional operation.

To evaluate the influence of the coupling effects and the inductance distribution, power losses are analyzed with respect to  $k$  and output load of the LDC, as illustrated in Fig. 9. The corresponding power profile and loss models are presented in Fig. 6(b) and Table V, respectively. It should be noted that the

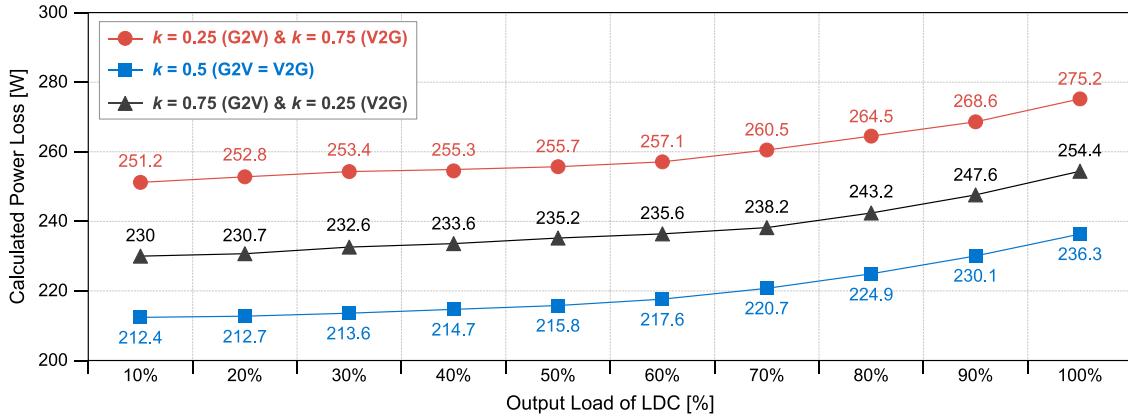


Fig. 9. Calculated power losses based on  $k$  ( $= L_T/L_P$ ) and output load of LDC at  $V_{Link} = V_{HV} = 650$  V,  $P_{Total} = 11$  kW, and  $L_T = 5$   $\mu$ H.

coupling effects and the inductance ratio collectively affect the overall performance.

Nonetheless, to isolate the effect of each factor,  $L_T$  was fixed throughout the analysis. Under this constraint,  $D_1$  remains constant when varying  $k$ , and therefore, asymmetry between the G2V and V2G modes can be attributed solely to the inductance ratio. Conversely, when  $L_T$  and  $k$  are fixed, variations in  $P_{LDC}$ , i.e.,  $D_1$ , result in variations in  $D_2$ . Accordingly, the impact of the coupling effects is evaluated by varying the LDC load while keeping  $k$  constant.

As illustrated in Fig. 9, the operation at  $k$  in the G2V mode corresponds to the operation at  $1-k$  in the V2G mode, since their input-to-output inductance ratios, i.e.,  $L_S/L_P$  in the G2V mode and  $L_P/L_S$  in the V2G mode, are matched under these respective conditions. Hence, deviations from  $k = 0.5$  result in asymmetric power losses in bidirectional operation. In addition,  $k = 0.5$  exhibits superior performance compared to other cases. This is because, under the same  $L_T$ , deviations from  $k = 0.5$  increase  $D_2$ , which in turn increases the switching and rms currents in the OBC switches, as indicated in Table IV, thereby degrading overall performance. Regardless of the inductance ratio, overall losses increase with  $P_{LDC}$ , and this is primarily attributed to the coupling effects. The reasons for this trend are discussed with experimental results in Section V.

Table VI summarizes the performance comparison based on  $k$ . While previous studies [15] and [16] selected  $k$  values lower than 0.5 without quantitatively analyzing the influence of the coupling effects and inductance ratio, this article selected  $k = 0.5$  to simultaneously reduce overall power losses and achieve symmetrical operation. Consequently, the proposed IBCS can improve the overall efficiency in both modes.

#### IV. TRANSFORMER DESIGN

This section presents the design process for the transformer parameters, including the turns ratio, magnetizing inductance, and leakage inductances, based on the specifications presented in Table II. Furthermore, the design criteria for core selection and a magnetic analysis of the designed transformer are presented.

TABLE VI

PERFORMANCE COMPARISON IN G2V AND V2G MODES AT UNITY GAIN

| $k$ ( $= L_T/L_P$ )    | $k < 0.5$  | $k = 0.5$ | $k > 0.5$ |
|------------------------|------------|-----------|-----------|
| $ D_2 $ (or $D_{C2}$ ) | G2V > V2G  | G2V = V2G | G2V < V2G |
| Efficiency             | G2V < V2G  |           | G2V > V2G |
| Design condition       | [15], [16] | Prop.     | -         |

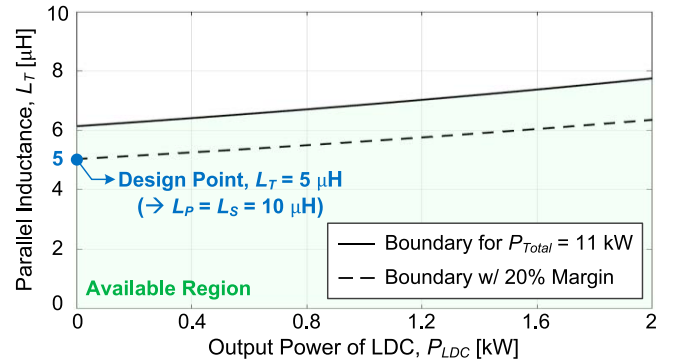


Fig. 10. Available region of  $L_T$  for  $P_{Total} = 11$  kW at worst case, i.e.,  $V_{Link} = 650$  V and  $V_{HV} = 500$  V.

#### A. Turns Ratio: $n$

The turns ratio  $n$  is designed considering the voltage gain of the LDC. The worst case occurs in the driving mode because, unlike the G2V and V2G modes,  $V_{Link}$  can be lower than 650 V. Therefore, based on (10), the following inequality is derived by assuming that the commutation interval accounts for a 10% reduction in the voltage gain and setting  $D_1 = 0$ :

$$n \leq \frac{(1 - D_1)V_{TM}}{2n} - \frac{L_T I_{LV}}{2n^2 T_{hs}} = \frac{0.9V_{TM}}{2V_{LV}}. \quad (13)$$

When  $V_{Link} = V_{HV} = 500$  V,  $V_{TM}$  becomes 500 V, as indicated in (3). However, in practice,  $V_{Link}$  is slightly lower than  $V_{HV}$  in the driving mode, as illustrated in Fig. 3(b). Considering this characteristic, an additional 10% design margin is applied to the case of  $V_{TM} = 500$  V, and the maximum available value of

TABLE VII  
CORE DESIGN PARAMETERS

| Parameters                                  | Value                                        |
|---------------------------------------------|----------------------------------------------|
| $L_{M0\text{OFF}11} (= L_{M2})/L_P (= L_S)$ | 1.3/10 $\mu\text{H}$                         |
| Turns ( $N_P, N_S, N_L$ )                   | 14, 14, 1 ( $n = 14$ )                       |
| Peak magnetic flux density                  | 0.3 T                                        |
| Wire current density                        | 600 A/cm <sup>2</sup>                        |
| Calculated $A_P$ for each core              | 155341 mm <sup>4</sup>                       |
| Selected core                               | Split PQ7651 ( $A_P = 252683 \text{ mm}^4$ ) |

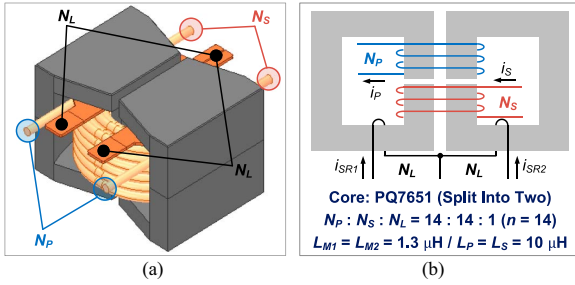


Fig. 11. Transformer design results. (a) Structure. (b) Cross-sectional view and parameters.

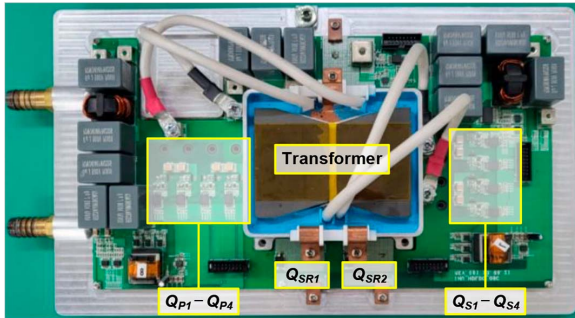


Fig. 12. Prototype of proposed IBCS.

$n$  is calculated to be approximately 14.46 from (13). Therefore, this article selects  $n = 14$ .

### B. Magnetizing Inductances: $L_{M1}$ and $L_{M2}$

In the two-transformer PSFB converter, the output current ripple decreases as  $L_{M1}$  and  $L_{M2}$  increase. However, due to an offset current of  $0.5I_{LV}$ , using large inductance values increases transformer volume. Considering this trade-off,  $L_{M1}$  and  $L_{M2}$  are designed to achieve a 40% output current ripple in the LDC at rated power. As illustrated in Fig. 3, the absence of  $D_2$  extends zero-voltage interval in  $v_T$ ; thus, the worst case occurs in the driving mode. Since both  $i_{M1}$  and  $i_{M2}$  reduce during  $(D_1 + D_C)T_{hs}$ , the output current ripple  $\Delta I_{LV}$  is derived as follows:

$$\Delta I_{LV} = (D_1 + D_C)T_{hs}V_{LV} \left( \frac{1}{L_{M1}} + \frac{1}{L_{M2}} \right) \leq 0.4I_{LV}. \quad (14)$$

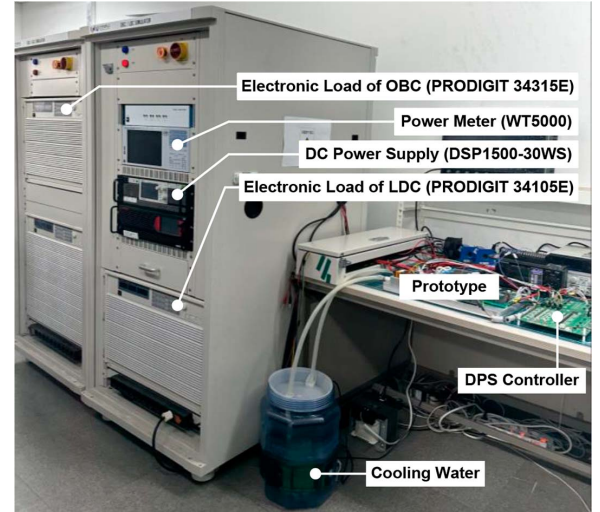


Fig. 13. Photograph of experimental setup.

By substituting (10) into (14) and  $L_{M1} = L_{M2} = L_M$ , the minimum value of  $L_M$  is calculated to be approximately 1.25  $\mu\text{H}$ . Based on this result,  $L_{M1}$  and  $L_{M2}$  are selected as 1.3  $\mu\text{H}$ .

### C. Leakage Inductances of OBC: $L_P$ and $L_S$

As indicated in Tables III and IV, the available  $L_T$  increases with  $D_2$  and is maximized when  $D_2 = D_1$ , where  $D_1$  can be obtained from (10). Therefore, based on the total power model, the following requirement for  $L_T$  can be derived:

$$L_T \leq \frac{V_{Link} T_{hs}}{8P_{Total}} [V_{HV} D_1 (2 - 3D_1) + V_{Link} D_{C2} (1 - D_1 - D_{C2})]. \quad (15)$$

Fig. 10 shows the available  $L_T$  for  $P_{Total} = 11 \text{ kW}$  at worst case, i.e.,  $V_{Link} = 650 \text{ V}$  and  $V_{HV} = 500 \text{ V}$ . As indicated in (10) and (15), the available  $L_T$  decreases as  $D_1$  increases, and  $D_1$  reaches its maximum under the no-load condition of the LDC, requiring  $L_T$  to be less than 6.2  $\mu\text{H}$ . Considering a 20% margin, this article selects  $L_T = 5 \mu\text{H}$ , resulting in  $L_P = L_S = 10 \mu\text{H}$ .

### D. Core Selection and Magnetic Analysis

Table VII summarizes design parameters for core selection. A PQ core is considered, and the peak magnetic flux density is limited to 0.3 T to prevent magnetic saturation. To minimize copper losses, the secondary winding of the LDC is fixed to a single turn, i.e.,  $N_L = 1$ , and the current density in all windings is restricted to 600 A/cm<sup>2</sup>. Accordingly, the turns for the OBC windings are determined as  $N_P = N_S = 14$ . Based on these parameters, the required area product,  $A_P$ , is calculated, and customized split PQ7651 cores are adopted. Compared to using smaller individual PQ cores, employing half-cores of a larger PQ core provides a greater window area for a given  $A_P$ , thereby reducing the overall core size. In addition, it facilitates the OBC windings to be commonly wound across both cores, effectively shortening the total wire length [10], [19]. The resulting transformer structure and parameters are shown in Fig. 11.

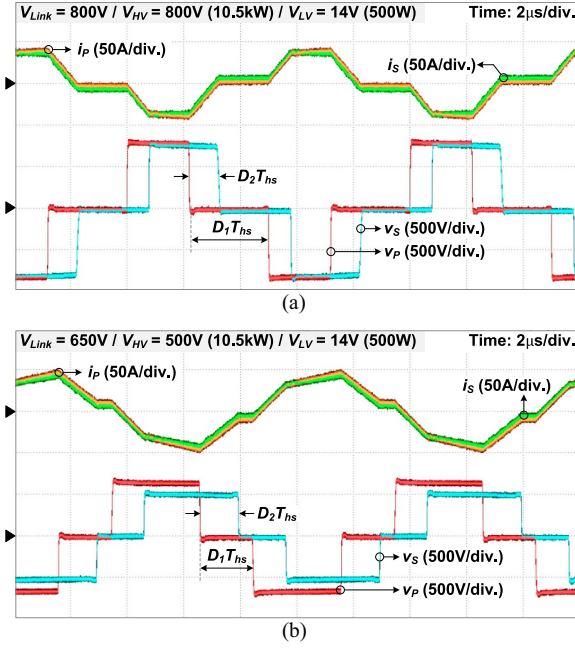


Fig. 14. Experimental waveforms in the G2V mode at  $P_{OBC} = 10.5$  kW and  $P_{LDC} = 500$  W, i.e.,  $P_{Total} = 11$  kW. (a)  $V_{HV} = 800$  V. (b)  $V_{HV} = 500$  V.

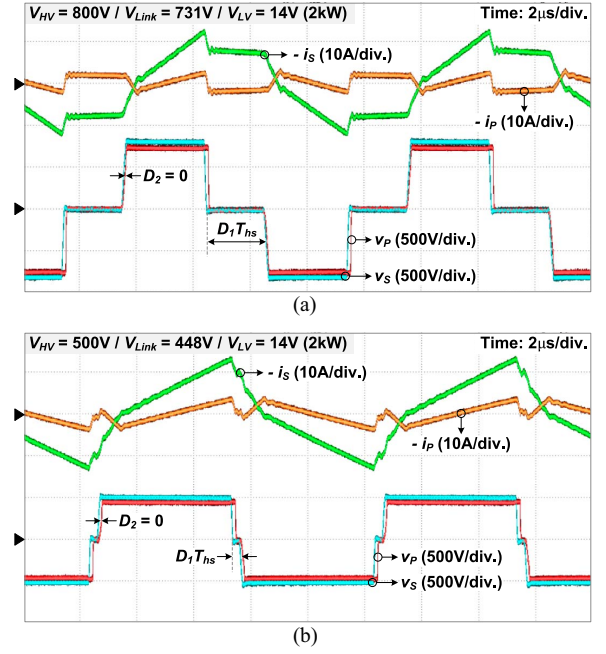


Fig. 16. Experimental waveforms in the driving mode at rated power, i.e.,  $P_{LDC} = 2$  kW. (a)  $V_{HV} = 800$  V. (b)  $V_{HV} = 500$  V.

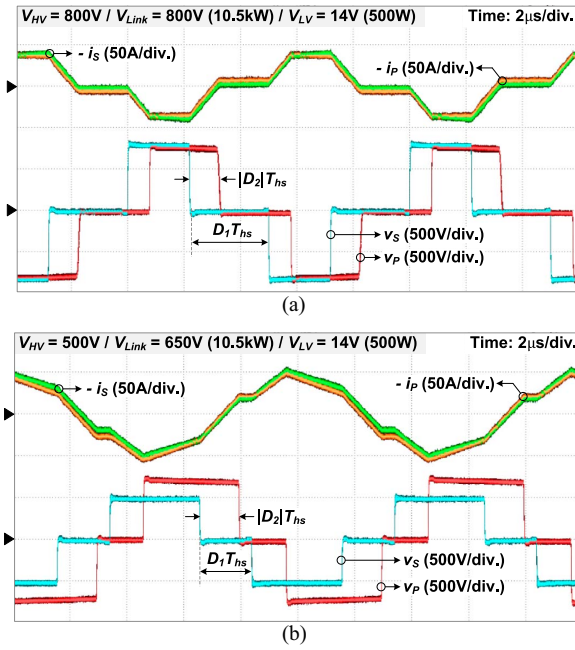


Fig. 15. Experimental waveforms in the V2G mode at  $P_{OBC} = 10.5$  kW and  $P_{LDC} = 500$  W, i.e.,  $P_{Total} = 11$  kW. (a)  $V_{HV} = 800$  V. (b)  $V_{HV} = 500$  V.

Each core shown in Fig. 11 is designed to have a magnetizing inductance of  $L_{M1}$  and  $L_{M2}$ , respectively. Assuming that the distance between split cores is sufficiently large, the magnetic flux linkage between them can be considered negligible. Under this condition, the following equations can be derived:

$$\phi_1 = L_{M1}[ni_P(t) - ni_S(t) - i_{SR1}(t)] \quad (16)$$

$$\phi_2 = L_{M2}[ni_P(t) - ni_S(t) + i_{SR2}(t)] \quad (17)$$

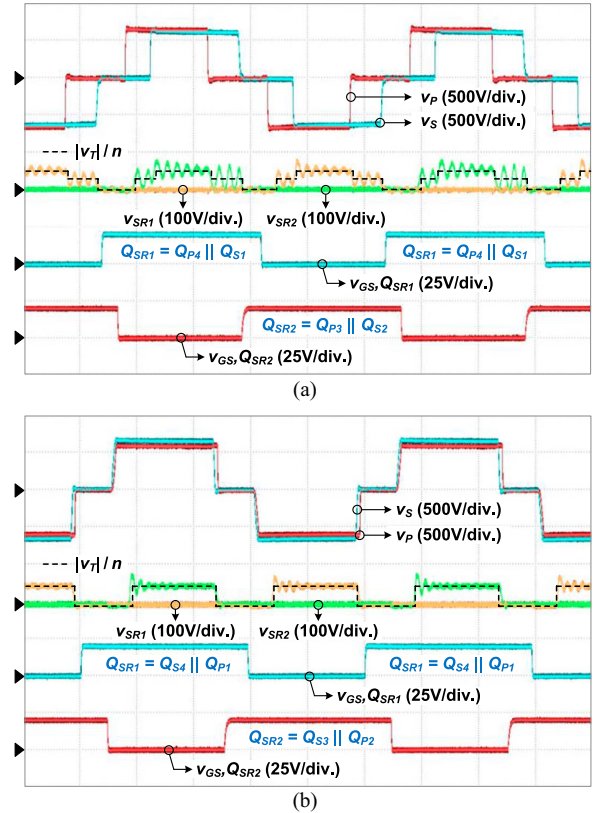


Fig. 17.  $v_T$  and gate signals for SRs in proposed method at  $V_{HV} = 650$  V. (a) G2V mode. (b) Driving mode.

where  $\theta_1$  and  $\theta_2$  represent the magnetic flux flowing through each split core. As illustrated in Figs. 2(b) and 3,  $n(i_P - i_S) - i_{SR1}$  and  $n(i_P - i_S) + i_{SR2}$  correspond to  $i_{M1}$  and  $i_{M2}$ , respectively. Therefore, by applying Faraday's law to (16) and (17), the

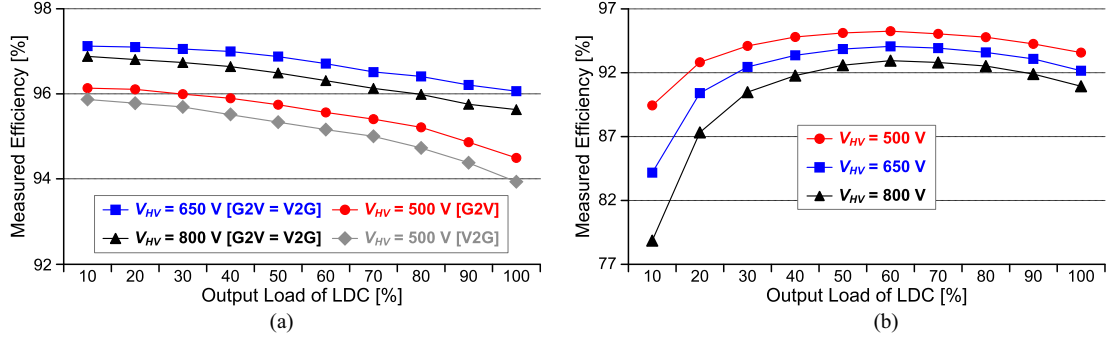


Fig. 18. Measured efficiency. (a) In G2V and V2G modes at  $P_{Total} = 11$  kW. (b) In driving mode.

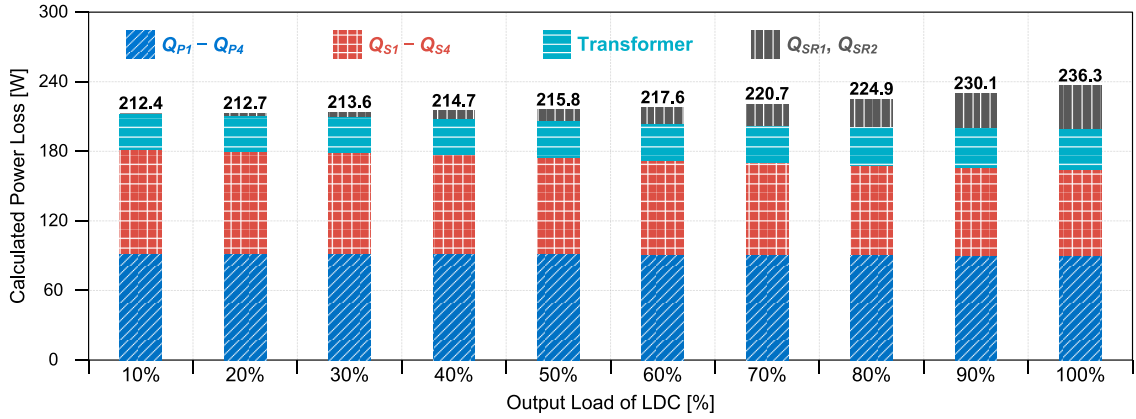


Fig. 19. Loss distribution in each component based on output load of LDC in G2V mode at  $V_{Link} = V_{HV} = 650$  V and  $P_{Total} = 11$  kW.

following equation is obtained:

$$v_P = \frac{N_P d(\phi_1 + \phi_2)}{dt} = \frac{L_{M1} di_{M1}(t)}{dt} + \frac{L_{M1} di_{M2}(t)}{dt}. \quad (18)$$

The circuit diagram of the proposed IBCS shown in Fig. 2(b) corresponds to the equivalent model defined in (18), which represents the magnetically decoupled structure. To ensure this condition, the two split cores were arranged at a sufficiently large distance, as shown in Fig. 11.

## V. EXPERIMENTAL RESULTS

To verify the performance of the proposed IBCS, a prototype integrating an 11 kW OBC and a 2 kW LDC was built, as shown in Fig. 12. Fig. 13 illustrates a photograph of the experimental setup, and the system specifications and transformer design results are summarized in Table II and Fig. 11. The MOSFETs of IMZ120R030M1H and IAUTN12S5N018T were used for the OBC switches and SRs, respectively. A TMS320F28379D DSP controller was utilized to control the proposed IBCS.

Figs. 14 and 15 show the key waveforms under rated power in the G2V and V2G modes, respectively. As illustrated in Figs. 14(a) and 15(a), when  $V_{Link} = V_{HV}$ , the proposed IBCS operates symmetrically. This is due to the equal design of  $L_P$  and  $L_S$ , as shown in Table VI, which enables simultaneous optimization of efficiency in both the G2V and V2G modes. In contrast, when the voltage gain of the OBC deviates from

unity, current slopes in States 1 and 2 differ in the G2V and V2G modes, as indicated in (1)–(5). Accordingly, under the minimum voltage conditions, i.e.,  $V_{Link} = 650$  V and  $V_{HV} = 500$  V, the waveforms become asymmetrical, as shown in Figs. 14(b) and 15(b).

Fig. 16 shows the key waveforms in the driving mode. As observed,  $D_2$  is set to zero, and  $D_1$  is varied with  $V_{HV}$  to regulate  $V_{LV}$ . Since the OBC primary side is under no-load conditions, only a small circulating current flows through this port. As mentioned in Section II-C, due to the commutation interval,  $V_{Link}$  is slightly lower than  $V_{HV}$ , showing approximately a 10% voltage reduction at rated power, as shown in Fig. 16(a) and (b). Considering this property, the turns ratio  $n$  was designed to provide sufficient margin. Accordingly, the proposed IBCS regulated  $V_{LV}$  across the entire battery voltage range.

Meanwhile, as illustrated in Figs. 14–16, the proposed IBCS operates under the operational constraints defined in Section III-C, thereby enabling the application of the proposed SR driving scheme presented in Table III. Fig. 17 shows the measured  $v_T$  and the gate signals of the SRs under the proposed method in the G2V and driving modes at  $V_{HV} = 650$  V, with power conditions identical to those in Figs. 14 and 16, respectively. Since  $v_T/n$  is reflected on the LDC secondary side, as shown in Figs. 4 and 5,  $v_T$  was indirectly measured from the drain-to-source voltages of  $Q_{SR1}$  and  $Q_{SR2}$ , i.e.,  $v_{SR1}$  and  $v_{SR2}$ . As shown in Figs. 3, 7, and 17, the measured waveforms closely align

with the theoretical analysis. Specifically, both SRs are turned ON during all zero-voltage intervals of  $v_T$ , which prevents body diode conduction of the SRs, thereby minimizing conduction losses on the LDC secondary side.

Fig. 18(a) illustrates the measured efficiency in the G2V and V2G modes. When  $V_{\text{Link}} = V_{\text{HV}}$ , the proposed IBCS achieves the same efficiency in bidirectional operation due to the equal design of  $L_P$  and  $L_S$ . Therefore, if  $V_{\text{Link}}$  can be adjusted to match  $V_{\text{HV}}$  throughout the entire operating range, the proposed IBCS can maintain high and identical efficiency. However, due to the minimum voltage limitation of the PFC converter [20], which is set to 650 V in this article,  $V_{\text{Link}}$  becomes higher than  $V_{\text{HV}}$  when  $V_{\text{HV}}$  falls below 650 V, as shown in Fig. 6(a). Thus, under the minimum voltage conditions, the efficiency difference in the G2V and V2G modes was observed, as shown in Fig. 18(a).

Fig. 18(b) shows the measured efficiency in the driving mode. As observed, the overall efficiency increased as  $V_{\text{HV}}$  decreased. This is primarily attributed to the reduction in switching and conduction losses. The switching losses in the switches  $Q_{P1}$ – $Q_{P4}$  and  $Q_{S1}$ – $Q_{S4}$  increase with  $V_{\text{Link}}$  and  $V_{\text{HV}}$ , respectively. In addition, a higher  $V_{\text{HV}}$  results in a higher  $D_1$ , which extends the zero-voltage intervals and increases conduction losses [6].

Fig. 19 shows the loss distribution in each component based on the LDC load in the G2V mode at  $V_{\text{Link}} = V_{\text{HV}} = 650$  V and rated power. The losses in  $Q_{S1}$ – $Q_{S4}$  decrease as  $P_{\text{LDC}}$  increases, while those in  $Q_{P1}$ – $Q_{P4}$  and the transformers remain nearly unchanged. This is because  $P_{\text{OBC}}$  decreases as  $P_{\text{LDC}}$  increases under the same total power condition. In contrast, the losses in the SRs increase with  $P_{\text{LDC}}$ , and this increase exceeds the reductions in other components. Thus, the overall power losses increase with  $P_{\text{LDC}}$ , showing the influence of coupling effects. This trend is consistently observed in the experimental results.

## VI. CONCLUSION

This article presented a new IBCS that combines the DAB converter for the OBC and two-transformer PSFB converter with the SRs for the LDC. Unlike conventional IBCSs, the proposed IBCS integrates the output inductors for the LDC into the transformers, thereby reducing the overall volume and the number of conducting components on the LDC secondary side. In addition, an optimal SR driving scheme was introduced to minimize conduction losses in the LDC, and the transformers were designed to optimize the efficiency of the proposed IBCS in both the G2V and V2G modes. These optimizations were based on a detailed analysis, including the coupling effects between the OBC and LDC and the influence of the leakage inductances on the OBC primary and secondary sides—factors not thoroughly addressed in previous studies. Experimental results validated the effectiveness of the proposed IBCS, confirming peak efficiencies of 97.1% in both the G2V and V2G modes at the unity voltage gain of the OBC, and 95% in the driving mode at the minimum voltage condition. It is worth noting that conventional IBCSs have not been designed to optimize bidirectional performance, and such efficiency has not been demonstrated under high-power conditions, particularly at or

above 11 kW. Therefore, the proposed IBCS is expected to be useful for high-power EV chargers.

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